

22. Perform Sharpening of Image using Laplacian mask with positive center coefficient.

Mask of Laplacian + addition

$$\begin{aligned}g(x, y) &= f(x, y) - [f(x+1, y) + f(x-1, y) \\&\quad + f(x, y+1) + f(x, y-1) + 4f(x, y)] \\&= 5f(x, y) - [f(x+1, y) + f(x-1, y) \\&\quad + f(x, y+1) + f(x, y-1)]\end{aligned}$$

0	-1	0
-1	5	-1
0	-1	0

AIM:

Sharpening of Image using Laplacian mask with positive center coefficient.

PROGRAM:

```
import cv2
import numpy as np

image = cv2.imread(r'C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png')

if image is None:
    print("Error: Image not found!")
else:
    # Laplacian sharpening kernel
    kernel = np.array([
        [0, 1, 0],
        [1, -8, 1],
        [0, 1, 0]
    ])

    # Apply the sharpening filter
    sharpened = cv2.filter2D(image, -1, kernel)

    # Save the output image
    cv2.imwrite(
        r'C:\Users\Susmitha\Downloads\ITA0510 Lab\Sharpened_Image.jpg',
        sharpened
    )

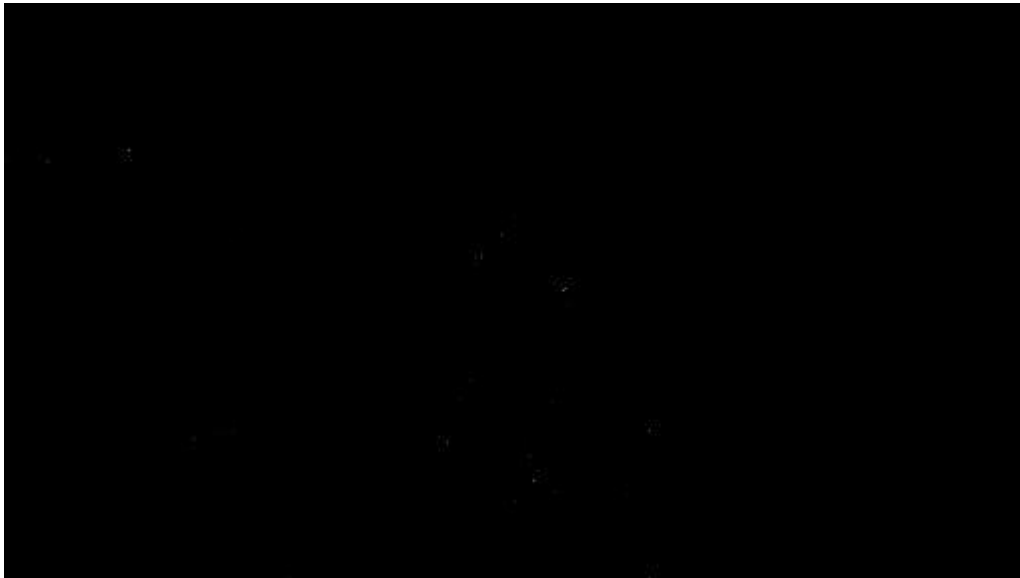
    # Display the images
    cv2.imshow("Original Image", image)
    cv2.imshow("Sharpened Image", sharpened)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

INPUT:



OUTPUT:



RESULT:

Thus the program is executed successfully.